

Let the agent see what it needs.

Dhristi · On-device Visual Perception for Light-weight Browser Agents

**Local perception.
A smaller privacy boundary.**

SIH26171
ISRO · Smart Automation
Software

A screen contains more than a task.

The same view can expose identity, account information, a password field and useful controls.

Sensitive context

Face · account · form
values

Useful structure

Geometry · approved
controls

Agent task

Find a pending request

[Problem basis: SIH26171. UI PII coverage is an active research problem: WebPII \(2026 preprint\).](#)

Optimize the real judging dimensions.

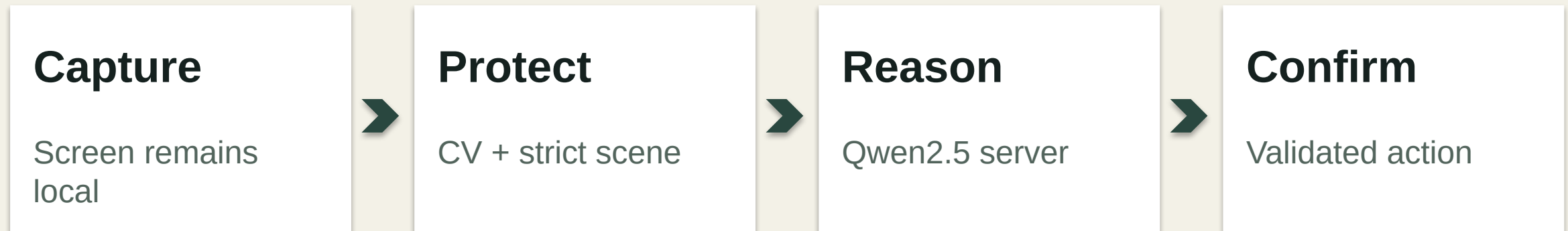
Organizer weights guide the evaluation plan. They are not our scores.



[Official weights: SIH 2026 catalogue. Complete scoring formulas are not supplied.](#)

The privacy decision stays in the browser.

Actual v0.1 flow: local CV + DOM → protected scene → server reasoning → reviewed local action.



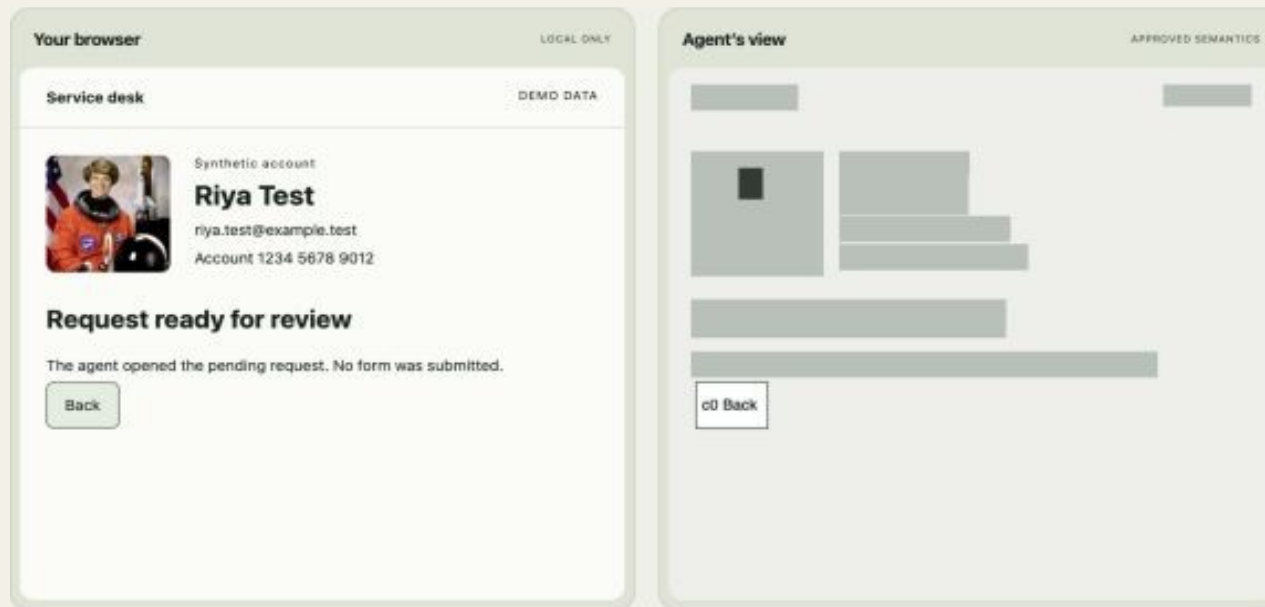
BROWSER · raw capture never enters the plan request

Only protected geometry / labels cross the API.

Implementation: Prototype/README.md · server LLM consumes layout data, not image pixels.

Inspect exactly what the agent receives.

Actual synthetic-demo capture. The protected view removes original text and media.



**Original screen
stays local.**

Approved controls
survive.

Unknown text and pixels
become opaque regions.

Screenshot: delivered local build, 9 Sep 2026. Synthetic account; NASA reference image, not team portrait.

Small model. Real local inference.

UltraFace RFB-320 runs in packaged ONNX Runtime Web 1.23.2, single-thread WASM.

MODEL INPUT

320 × 240

RGB → (value – 127) / 128
NCHW float32 tensor

ASSETS · DECIMAL MB

1.26 MB

Optimized model bytes
Runtime adds ≈11.98 MB

DETECTION LIMIT

Face-only

Not OCR or general PII
Errors block preparation

[Model/runtime manifest records exact bytes, hashes and MIT licenses. No browser CDN.](#)

A typed boundary is easier to audit.

The outgoing plan request has an allowlist; original screen content has no payload field.

ALLOWED CONTEXT

```
revision + viewport  
approved control IDs /  
labels  
bounded geometry  
opaque region kinds
```

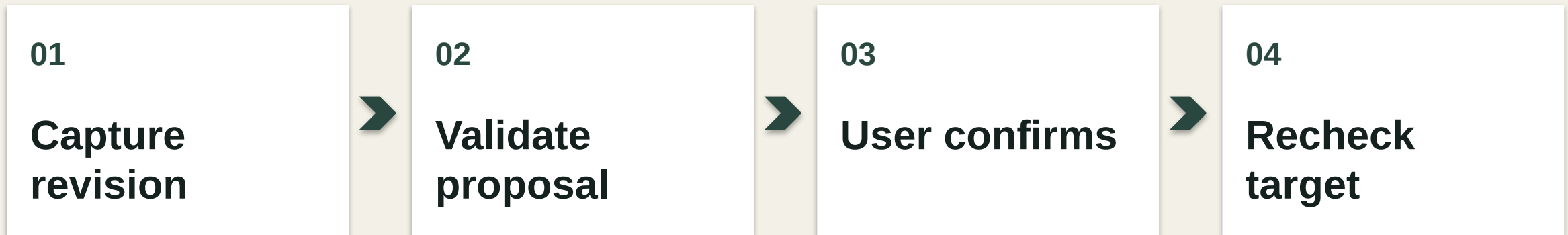
Excluded

Raw screenshots / DOM
Account and field values
Source URLs / arbitrary text
Credentials

Observed payload excluded fixture email, account number and image data at 390px. One synthetic fixture.

A model proposal is not execution authority.

Every action must match the live scene and survive local validation.



Allowed: click / scroll / done. No arbitrary URL, typing, script or submission.

30-second capture expiry; delayed action rejected in the recorded workflow. Prompt-injection safety is not complete.

One complete task, with a real model.

Observed synthetic workflow: open Pending, then Review. No external account or form submission.

Requests

Capture → Qwen → confirm

Pending requests

Recapture → Qwen →
confirm

Ready for review

Correct fixture end-state

Demo fallback: local screenshots available. Screen-recording playback remains a separate delivery check.

Recorded manual integration: actual qwen2.5:7b-instruct on Ollama, not provider test doubles.

Fast perception. Slower reasoning.

Single integration observations on Apple M1 Pro / 16 GB / embedded Chromium. Not p95.

9–20 ms

Face inference, 14 input
sizes

92–108 ms

Capture + protection

3,029 ms

First real model step

119 recorded Node tests pass (2026-09-15). The <200 ms full-flow target does not pass.

Face-inference range: 11 Sep 2026 face-scale matrix. Capture and model figures: 9 Sep 2026 v0.1 record. Test doubles are unit-test scope; the Qwen workflow is separate evidence.

Build on prior work, then test the difference.

Privacy filtering is established. Our claim is an inspectable browser integration, not invention of anonymization.

WebPII / WebRedact	UI PII detection + benchmark	Browser cost and action-boundary tests pending
PrivWeb	Local anonymization + user control	Explicit pixel-free egress + expiring controls
Available but Invisible	Typed placeholders + secure proxy	Browser implementation and packaging

[Primary preprints: arXiv 2603.17357 · 2509.11939 · 2602.10139. No head-to-head accuracy study.](#)

The strict boundary has a utility cost.

Removing unknown text and images is conservative. It can also remove the answer.

CURRENT

Known limits

Face-only model
Restricted task vocabulary
LLM reads layout, not PNG

OPEN

Validate next

Chrome + Firefox extension
Labeled unseen pages
Accessibility and user review

ROADMAP

Improve carefully

Compare DOM / CV baselines
Worker and cold-load profile
Add context only with evidence

These are explicit limitations and planned work. No broad PII, WCAG or extension conformance claim.

Measure benefit at the task boundary.

Proposed pilot outcomes, not achieved social impact or a validated business model.

User control

Comprehension of preview
Review burden
Task completion

Operating viability

Existing-device deployment
Cost per completed task
Maintainer responsibility

Privacy / utility

Sensitive coverage
Useful context retained
Failure severity

A local-install institutional pilot is a proposed route. No partner, revenue, savings or energy benefit is asserted.

Six owners. One accountable demo.

Responsibilities are proposed work allocation; no individual experience or awards are invented.

Francis

Integration + technical pitch

Gopreet

API + persistence

Hiranmayi

UX + accessibility

Varun

Domain implementation

Koushaik

Testing + reliability

Niharika

Research + narrative

09 SEP Build evidence → 10 SEP Validate + rehearse → 11 SEP Internal delivery

User-supplied team names and institution. Registered team name/ID and consented portraits remain pending.

Useful context. A smaller exposure.

Dhristi demonstrates a reviewed browser-agent loop; evaluation determines how far it can go.

Next decision: test the boundary on held-out tasks.

Inspect the context. Challenge the action. Measure the trade-off.

Run locally: [Prototype/README.md](#) · Evidence: [Benchmarks/results/](#) · Source ledger: [Wiki/source-index.md](#)